Minimal persistence grids and homological Morse bounds

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Abstract

The persistent homology of a finite simplicial complex is determined by a canonical grid whose coordinate values occur in minimal generators and relations. We prove that the number of values in each coordinate is bounded by the homological Morse lower bound defined from the multigraded Betti tables. The result holds over every field, for every number of parameters, and for filters with coincident coordinate values. It establishes the conjecture in Remark 5.12 of Harrington, Tillmann and Torras-Pérez. The proof selects a minimal presentation degree on each coordinate slice. At that degree a local Koszul calculation eliminates higher syzygies; injectivity of top-dimensional homology handles the final homology degree.

1 The grid and the lower bound

Persistent homology records the homology of sublevel complexes together with the maps induced by their inclusions. In the inverse problem, one asks which filters give the same persistent homology. Harrington, Tillmann and Torras-Pérez [1] obtain dimension estimates for these spaces of filters using both a minimal determining grid and a lower bound for homological Morse numbers. Their Remark 5.12 asks whether each coordinate of the minimal grid has cardinality at most that lower bound. We prove this comparison for general filters, including coordinate coincidences.

Fix a field $\Bbbk$, an integer $n \geq 1$, and a finite nonempty simplicial complex K of dimension d. A filter assigns $f(\sigma) \in [0,1]^n$ to every nonempty simplex, with $f(\tau) \leq f(\sigma)$ coordinatewise whenever $\tau \subseteq \sigma$. For $u \in \mathbb{R}^n$, ordered coordinatewise, set

$$K_u = \{\sigma \in K : f(\sigma) \leq u\}, \qquad M_u^q = H_q(K_u; \Bbbk), \qquad 0 \leq q \leq d,$$

using ordinary unreduced homology and inclusion-induced maps. Write $\mathbf{M} = (M^0, \dots, M^d)$. A free summand born at a grade w has value $\Bbbk$ at every $v \geq w$ and zero elsewhere; all maps between its nonzero components are identities. Let $\xi_i^q(u)$ be the multiplicity of a free summand at grade u in position i of a minimal free resolution of M^q. We set these numbers to zero when $q < 0$ or $q > d$.

Let

$$S = \bigcup_{q=0}^{d} \operatorname{supp}(\xi_0^q + \xi_1^q), \qquad G_j = \{s_j : s \in S\}, \qquad G = \prod_{j=1}^{n} G_j. \tag{1}$$

By [1, Lemma 1.23 and Definition 2.9], G is the canonical minimal grid determining the tuple. Values 0 and 1 are included only when they occur in the indicated supports. The lower bound in

question is

$$b_q(u) = \max\left\{0, \xi_0^q(u) + \xi_1^{q-1}(u) - \sum_{p=1}^{n-1} \xi_{p+1}^{q-p}(u)\right\}, \tag{2}$$

$$\mathcal{L}(\mathbf{M}) = \sum_{u\in G}\sum_{q=0}^{d} b_q(u). \tag{3}$$

The expression in (2) comes from the Morse inequalities of Guidolin and Landi [2, Theorem 7.3]; (2)–(3) are the quantities used in [1, Definition 5.7 and Proposition 5.9]. Their grid comparison is known for one parameter and for filters injective in every coordinate [1, Lemma 5.10 and Remark 5.12]. For general filters the higher-syzygy terms can cancel the positive terms at some grades. We will find a grade with a positive contribution on every coordinate slice of G.

We use consecutive integer indices for each ordered G_j and extend the modules along this grid to $\mathbb{Z}^n$. This regrading preserves minimal free resolutions and their summand multiplicities [1, Section 1.4]. Each coordinate set keeps its cardinality. In these indices put $R = \Bbbk[x_1, \dots, x_n]$, with $\deg x_j = e_j$, where e_j is the jth standard basis vector. A free summand $R(-w)$ has a copy of $\Bbbk$ at v precisely when $w \le v$, and its maps between present components are identities. Each module has a minimal presentation

$$F_1^q \longrightarrow F_0^q \longrightarrow M^q \longrightarrow 0, \tag{4}$$

whose basis degrees are counted by ξ_1^q and ξ_0^q. The common grid determines every component, even if a component has a smaller minimal grid. Below its support the extended module is zero.

2 A local Koszul calculation

For a finitely presented integer-graded R-module V and a grade u, its Koszul complex is

$$\mathbb{K}_i(V)(u) = \bigoplus_{\substack{\alpha\subseteq\{1,\dots,n\}\\ |\alpha|=i}} V_{u-e_\alpha}, \qquad e_\alpha = \sum_{k\in\alpha} e_k.$$

The differential is the alternating sum of the coordinate maps that delete one index from α. Its homology computes the Betti tables:

$$\xi_i(u) = \dim_{\Bbbk} H_i(\mathbb{K}_*(V)(u)); \tag{5}$$

see the integer-graded derivation in [1, Section 1.4], followed by Definition 1.30 and Lemma 1.31 there. The next lemma needs isomorphisms in just one coordinate, away from the vertex u.

Lemma 1. *Fix $j \in \{1, \dots, n\}$. Suppose that*

$$V_{u-e_j-e_\alpha} \longrightarrow V_{u-e_\alpha} \tag{6}$$

is an isomorphism for every nonempty $\alpha \subseteq \{1, \dots, n\} \setminus \{j\}$. Then $\xi_i(u) = 0$ for all $i \ge 2$. If, in addition, $V_{u-e_j} \to V_u$ is injective, then $\xi_1(u) = 0$.

Proof. Let A and B be the Koszul complexes in the variables other than x_j, at grades $u - e_j$ and u, respectively. Multiplication by x_j gives a chain map $\phi : A \to B$. The hypothesis says that ϕ_p is an isomorphism for every $p \ge 1$, including zero terms above the lengths of these complexes.

Choose the exterior ordering with j first. Separating the subsets that contain j identifies the full Koszul complex with

$$C_i = B_i \oplus A_{i-1}, \qquad D_i(b,a) = (d_B b + \phi_{i-1} a, -d_A a).$$

The minus sign comes from differentiating an exterior product $e_j \wedge \eta$: its two terms are $x_j \eta - e_j \wedge d\eta$.

For a cycle $(b,a) \in C_i$ with $i \geq 2$, take $c = \phi_i^{-1} b$. The cycle equation and $d_B \phi = \phi d_A$ give

$$\phi_{i-1}(a + d_A c) = d_B b + \phi_{i-1} a = 0.$$

Since ϕ_{i-1} is injective, $a = -d_A c$. Thus

$$(b,a) = D_{i+1}(0,c)$$

is a boundary, proving the first assertion by (5).

For a degree-one cycle, the same choice $c = \phi_1^{-1} b$ reduces it, modulo the boundary $D_2(0,c)$, to $(0, a + d_A c)$ with $\phi_0(a + d_A c) = 0$. Injectivity of ϕ_0 makes that cycle zero. When $n = 1$, A and B have only degree-zero terms, so the assertion is the kernel calculation for $V_{u-e_j} \to V_u$. □

3 The cardinality bound

The last homology degree requires a simple fact about the common underlying complex. For every inclusion $K_v \subseteq K_w$, the map $H_d(K_v; \Bbbk) \to H_d(K_w; \Bbbk)$ is injective: there are no $(d+1)$-simplices, so these homology groups are the kernels of the degree-d boundary, and the chain map includes one kernel in the other. At $d = 0$, ordinary unreduced homology has $\partial_0 = 0$, and the same argument applies. Thus all coordinate maps of M^d are injective, also after the grid regrading.

Theorem 2. *Let K be any finite nonempty simplicial complex, $\Bbbk$ any field, and $n \geq 1$. For every face-monotone filter $f : K \to [0,1]^n$, including filters with coincident coordinate values, its ordinary unreduced persistent homology tuple satisfies*

$$|G_j| \leq \sum_{u \in G} \sum_{q=0}^{d} \max \left\{ 0, \xi_0^q(u) + \xi_1^{q-1}(u) - \sum_{p=1}^{n-1} \xi_{p+1}^{q-p}(u) \right\} = \mathcal{L}(\mathbf{M}) \quad (1 \leq j \leq n).$$

Here $d = \dim K$, ξ_i^q are the multigraded Betti tables of $H_q(K_u; \Bbbk)$ from this same filter, with absent homology degrees assigned zero, and $G = \prod_j G_j$ is the canonical minimal grid defined in (1).

Proof. Fix j and $t \in G_j$. In the finite nonempty set $\{s \in S : s_j = t\}$ choose an element u minimal in the coordinatewise order. There is consequently no $s \in S$ with $s_j = t$, $s \leq u$ and $s \neq u$. This choice uses the presentation degrees from all homology degrees together.

For a nonempty $\alpha \subseteq \{1, \ldots, n\} \setminus \{j\}$ put $v = u - e_\alpha$. No basis degree w of either F_0^q or F_1^q can satisfy $w \leq v$ and $w_j = t$, for such a w would belong to S and be strictly below u on the same slice. All free summands present at v therefore have $w_j \leq t - 1$, and are already present at $v - e_j$. Multiplication by x_j gives isomorphisms on both free modules of (4) at these two grades. Their commuting presentation maps identify the images, hence the cokernels. We obtain

$$M^q_{u-e_j-e_\alpha} \xrightarrow{\cong} M^q_{u-e_\alpha} \qquad (0 \leq q \leq d).$$

Lemma 1 yields

$$\xi_i^q(u) = 0 \quad (i \geq 2,\ 0 \leq q \leq d).$$

Its last assertion, applied to the injective coordinate map of M^d, also gives $\xi_1^d(u) = 0$.

Since $u \in S$, some $\xi_0^q(u)$ or $\xi_1^q(u)$ is positive. In the latter case $q < d$. At this selected grade all subtraction terms in (2) vanish, so

$$b_q(u) = \xi_0^q(u) + \xi_1^{q-1}(u), \qquad \sum_{q=0}^{d} b_q(u) \geq 1.$$

Each selected $u = u(t)$ belongs to G. Distinct values of t give distinct grades because $u(t)_j = t$. Nonnegativity of every b_q now gives

$$\mathcal{L}(\mathbf{M}) \geq \sum_{t \in G_j} \sum_{q=0}^{d} b_q(u(t)) \geq |G_j|.$$

This proves the result for each coordinate. An empty coordinate set would give the inequality $0 \leq \mathcal{L}(\mathbf{M})$ directly. □

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References

[1] H. A. Harrington, U. Tillmann and M. Torras-Pérez, *The fiber of multiparameter persistent homology for simplicial complexes*, preprint, 2026, arXiv:2608.04640v1. https://arxiv.org/abs/2608.04640v1.

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